

Quick Check

Lesson 2: What Is Pressure? Measuring the Invisible Push — Shoes, Scales, and Force Per Unit Area

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.1: Pressure | Lesson 2

A sealed empty plastic jerrycan has had some air removed from inside and it is crumpling inwards. Which statement best describes how the air OUTSIDE is acting on it?

- A** It pushes only on the sides, because the top and bottom are flat
- B** It pushes only downwards on the top of the jerrycan
- C** It pushes inwards on every face: top, bottom and all four sides
- D** It does not push at all; the plastic simply becomes weak

A teacher presses a flat textbook onto her palm, then presses a sharp pencil tip onto her palm with about the same force. The pencil feels much more painful. What is the best explanation?

- A** The book has a smaller area, so it spreads the force less
- B** The force is about the same, but the pencil tip has a much smaller contact area, so the pressure is greater
- C** The pencil is heavier than the book
- D** The pencil tip pushes with a much bigger force than the book

A student has a mass of 50 kg ($g = 10 \text{ N/kg}$). She stands on one shoe whose sole has a contact area of 200 cm^2 . Taking her weight as shared equally between both feet, what pressure does one shoe exert on the floor? ($1 \text{ m}^2 = 10,000 \text{ cm}^2$)

A 1.25 Pa

B 25,000 Pa

C 12,500 Pa

D 125,000 Pa

Which choice correctly gives the definition of pressure and its SI unit?

- A** Pressure = area $\div$ force; unit is the Newton (N)
- B** Pressure = mass $\div$ area; unit is the kilogram (kg)
- C** Pressure = force $\times$ area; unit is the square metre (m²)
- D** Pressure = force $\div$ area; unit is the Pascal (Pa), which is N/m²

A maize bag of mass 90 kg ($g = 10 \text{ N/kg}$) is placed on a wooden pallet in a warehouse in Eldoret. The pallet has 4 legs, each with a base area of 25 cm^2 . The weight is shared equally by the legs. What pressure does each leg exert on the floor? ($1 \text{ m}^2 = 10,000 \text{ cm}^2$)

A 90,000 Pa

B 9,000 Pa

C 900,000 Pa

D 360,000 Pa

A Maasai elder leans on a rungu whose handle tip has an area of 2 cm^2 , with a force of 150 N . A nurse at Kenyatta National Hospital leans on a flat crutch base of area 50 cm^2 with the same 150 N force. How do the pressures on the floor compare?

- A The rungu exerts 48 times more pressure than the crutch
- B The pressures are equal, because both forces are 150 N
- C The rungu exerts 25 times more pressure than the crutch
- D The crutch exerts 25 times more pressure than the rungu

Which statement best describes atmospheric pressure?

- A** It is a force that acts only downwards on the top of objects
- B** It is the weight of the column of air above a surface, and it acts on every exposed face of an object
- C** It is the force of moving wind, and it acts only in the direction the wind blows
- D** It is the pressure of the air trapped inside a container only

An empty sealed jerrycan with normal air inside and outside stays intact. After some air is pumped out of it, it collapses. Which explanation is correct?

- A** Before, there was no air pushing on the jerrycan at all
- B** Before, the air pressure inside pushing outwards balanced the air pressure outside pushing inwards; after, the pressure inside was much less, so the outside pressure crushed it
- C** The air outside became heavier after the air inside was removed
- D** The plastic was already weak and only needed a small push from the pump

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).